Ian Kolaja

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Summary

Nuclear engineer with experience in reactor physics, core design, and nuclear data analysis for advanced reactors. Skilled in high-fidelity neutronics Monte Carlo and depletion modeling (Serpent, MCNP), gamma spectrometry modeling (GADRAS), and machine learning-based reactor diagnostics and control using neural networks. Demonstrated success developing core simulators and data-driven models for pebble bed and molten salt reactors. Strong background in computational methods, cross section generation and analysis, and reactor operations support.

Education

University of California, Berkeley — *August 2021 to August 2025*

PhD, Nuclear Engineering — *3.63*

with a Designated Emphasis in Computational and Data Science and Engineering

Highlighted Courses: Applications of Parallel Computers, Graduate Nuclear Reactions and Interactions of Radiation with Matter, Nuclear Security: The Nexus Between Policy and Technology, Advanced Nuclear Reactors, Special Topics in Environmental Aspects of Nuclear Energy: Thermal Hydraulics, Applied Machine Learning, Advanced Concepts in Radiation Detection and Measurements, Physical Principles of CT, PET, and SPECT Imaging, Introduction to Statistical Computing, Concepts of Probability

University of California, Berkeley — *August 2017 to December 2019*

B.S., Nuclear Engineering — *3.79*

Highlighted Courses: Nuclear Criticality Safety, STEM Communication, Radiation Biophysics and Dosimetry, Design for Advanced Nuclear Systems

Fullerton College — *August 2015 to June 2017*

A.S., Mathematics and Chemistry — *3.76*

Work Experience

Kairos Power — *June 2019 to September 2021*

Core Design Engineer

- Developed KPACS, a Python-based pebble bed reactor simulator, that wraps around Serpent to perform depletion and transport calculations. KPACS takes in a user-defined operation sequence and tracks the average motion of pebbles through the core, providing high-fidelity reactor data including reactivity, kinetics, flux distributions, reactivity coefficients, and fission product inventories by zone and number of passes. Achieved more accurate depletion results about 11 times faster than the previous method.
- Built a Python tool for automated Serpent model generation, representing reactor components as Python objects. Automated LaTeX-based technical report generation to ensure consistent documentation of model details.
- Verified core mixtures remained within safe operating limits by calculating reactivity coefficients. Ensured that the power per pebble did not exceed fuel qualification limits. Performed both infinite lattice calculations and full core simulations.
- Established GitHub repositories and comprehensive documentation for all developed tools, including user manuals and in-code commentary. Conducted hands-on training with colleagues, ensuring long-term usability after my departure.

UC Berkeley — *August 2021 to August 2025*

Graduate Student Researcher

- Inventor on provisional patent for a bent crystal diffraction (BCD) spectrometer system that can filter gamma rays by energy, enabling more rapid and accurate assessment of fuel burnup and plutonium production in continuously fueled reactors without excessive fuel decay time.
- Generated gamma spectra for pebble bed reactors (PBRs) via pebble-wise depletion calculations in Serpent. Simulated BCD spectrometer measurements using SHADOW3 ray tracing and GADRAS gamma spectroscopy.
- Extended methodology to molten salt reactors (MSRs), performing unit cell depletion in liquid fuel and simulating corresponding gamma spectra.
- Developed machine learning (ML) model to predict pebble-wise burnup, fluence, residence time, and nuclide content from measurable gamma spectra. Achieved R^2 of 0.994 for burnup prediction.
- Built PBR core simulator using Serpent that tracks zone-wise pebble composition data, enabling fast and accurate prediction of keff and flux/power distribution in the core as a function of various operator control inputs.
- Trained a Long-Short Term Memory (LSTM) neural network to predict reactivity and flux/power profiles using input features derived from measurable quantities. Performed hyperparameter tuning using K-fold cross validation. Achieved R^2 of 0.995 for reactivity on testing data and enabled forecasting of reactivity.
- Designed a control algorithm that uses LSTM forecasts to steer the reactor simulator from towards user-defined target states (i.e. full power) while maintaining criticality. Applied method to optimize running-in phase of PBR.
- Prototyped a flask-based web application in Python to assess and visualize the quality of nuclear cross section data from EXFOR and ENDF, incorporating metrics like uncertainty, experimental consistency, and spectral relevance. Built a user-facing interactive chart of nuclides for browsing nuclide and reaction specific scores.

UC Berkeley — *January 2022 to May 2022*

Graduate Student Instructor (GSI)

- Taught Introduction to Nuclear Reactor Theory, a core upper division course in Nuclear Engineering.
- Held hybrid discussion sections going over course material in greater detail with an emphasis of helping students understand how to utilize content covered in lectures. Organized review sessions and held office hours to ensure students were prepared.
- Received a 4.94 out of 5.0 on various student-graded metrics including teaching effectiveness, helpfulness in understanding course material, and preparedness.

Lawrence Berkeley National Lab *January 2018 to December 2019*

Undergraduate Researcher

- Determined a physically justifiable neutron flux spectrum for a fast neutron inelastic scattering experiment conducted at the Baghdad Nuclear Research Institution in 1978. Performed sensitivity analysis on the fitted flux function parameter in order to determine its uncertainty. Ongoing work will allow for the data coupled with the flux spectrum to serve as a nuclear model performance benchmark.
- Assessed the performance of the neutron flux spectrum by using it in a comparison between the experimental data and cross section calculations performed by TALYS, a nuclear reaction model.
- Developed code for the automatic analysis of gamma spectra that can identify and fit peaks using radioactive decay data. Utilized code to rule out the presence of fission products in the analysis of actinium-225, a coveted medical isotope.
- Acquired data for detection of super heavy element moscovium-288 using the Berkeley Gas-filled Separator (BGS) coupled with the 88-inch cyclotron. Utilized LabVIEW software to monitor detection events and adjust BGS parameters over time.

UC Berkeley Thermal Hydraulics Lab *November 2018 to December 2019*

Undergraduate Researcher

- Developed software in MATLAB and LabVIEW allowing a MIDI controller to be used as an analog controller for the Compact Integral Effects Test (CIET) facility.
- Developed an electronic procedure module in LabVIEW that guides reactor operators through experiments in CIET. Designed a data structure that can be easily created by other researchers to dynamically populate the electronic procedure with human readable instructions. The module is responsive to CIET's state, automatically advancing through the procedure as certain physical conditions are met.
- Designed procedures with graduate student for human factors experiment intended to test the effectiveness of different operator support tools. Conducted survey to supplement experimenter observations with participant's user feedback.

UC Berkeley *August 2018 to December 2018*

Course Reader

- Graded homework assignments and examinations for Nuclear Reactions and Radiation course at UC Berkeley. Contributed to the development of assignment and exam content using insight as a student. Hosted office hours to answer student questions about course material and grading policy.

Community Service

JOYA Scholars *September 2015 to May 2017*

Tutor and Mentor

- Tutored ten high school students from families of low-income communities to help a population that historically has had little access to higher education excel academically. Helped students with math, computer science, and biology.
- Mentored five middle school students in career selection, college admission, and overcoming academic challenges. Inspired one student to pursue aerospace engineering as a major and apply to Stanford University.

Fullerton College Associated Students *September 2015 to February 2016*

Student Senator

- Served as vice chair of Education and Curriculum Committee. Planned events augmenting student success. Among most volunteer hours served in Senate.
- Served on the district's Council on Budget and Facilities and Men and Women of Distinction Committee. Assisted in drafting of resolutions advocating for students who become homeless during the semester and increasing health fees to improve mental health resources on campus.

Skills

- **Programming Languages & Version Control:** Python, MATLAB, C++, R, Lua, Git, Conda
- **Scientific Modeling & Simulation:** Serpent, MCNP, TALYS, SHADOW 3, GADRAS, RELAP5, ORIGEN
- **Data Science & ML Tools:** Keras, scikit-learn, High Performance Cloud Computing, SLURM, Python Multiprocessing, CUDA
- **ML Techniques:** Regression, Classification, Clustering, Neural Networks (Feed forward, convolutional, recurrent), PCA, Time Series Forecasting, Feature Importance Analysis, K-folds Cross Validation
- **Web & UI Development:** HTML, CSS, JavaScript, Flask, Apache2, SQLite
- **Scientific Visualization:** Matplotlib, Bokeh, Mayavi
- **Instrumentation:** Gamma Spectroscopy, LabVIEW
- **Documentation & Communications:** Microsoft Office Suite, LaTeX, Overleaf, Speech Writing, Technical Writing, Social Media Outreach
- **Operating Systems:** Linux, Windows, MacOS
- **Creative & Media Tools:** Adobe Premiere Pro, Adobe Photoshop, Music Production, Ableton, Music Theory

Preprints

- I. Kolaja et al., "Predicting and forecasting reactivity and flux using long short-term memory models in pebble bed reactors during run-in," *arXiv*, November 2025, doi: https://arxiv.org/abs/2511.05118
- I. Kolaja et al., "Burnup measurement using bent crystal diffraction spectrometers for pebble bed reactors," *arXiv,* **(Submitted to journal)**, October 2025, doi: https://arxiv.org/abs/2510.08835

Publications

- I. Kolaja et al., "Machine Learning Prediction of Pebble History in Pebble Bed Reactors," *Proceedings of the International Conference on Physics of Reactors (PHYSOR 2024, pp. 1437-1446)*, April 2024, doi: 10.13182/PHYSOR24-43890
- N. Satval et al., "Neutronics, thermal-hydraulics, and multi-physics benchmark models for a generic pebble-bed fluoride-salt-cooled high temperature reactor (FHR)," *Nuclear Engineering and Design, vol. 384, no. C*, November 2021, doi: 10.1016/j.nucengdes.2021.111461
- J. M. Gates et al., "First Direct Measurements of Superheavy-Element Mass Numbers," *Phys. Rev. Lett. 121, 222501*, November 2018, doi: 10.1103/PhysRevLett.121.222501

Awards

Outstanding GSI *March 2023*

UC Berkeley Graduate Division Teaching & Resource Center

Awarded in recognition for exceptional achievements as a teacher. About ten percent of GSIs teaching on UC Berkeley's campus receive this award.

Wootton Award for Superior Service and Leadership *June 2020*

Berkeley Nuclear Engineering

Awarded in recognition of superior service and leadership to the Department of Nuclear Engineering

Chancellor's Fellowship for Graduate Study *March 2020*

Berkeley Nuclear Engineering

Nominated by department faculty and selected after rigorous review by a campuswide faculty committee in a highly competitive process. Funded two years of PhD study at UC Berkeley.

U.S. Nuclear Regulatory Commission Scholarship *May 2018*

NRC

Awarded by NRC for commitment to the field and remaining above a 3.0 GPA.

Student of Distinction *May 2017*

Fullerton College

Selected among twenty students recognized for academic achievement in a school of twenty-five thousand students by committee of faculty and students. Invited to audition to speak at commencement, became one of four finalists.

Carbon Contributions Challenge Finalist *March 2017*

MIT Solve

Proposed a solution with team for Massachusetts Institute of Technology (MIT) Solve's Carbon Contributions Challenge and was selected as one of eleven finalists. Solutions were accepted from around the world with many finalists being graduate students from top tier schools. Designed a process called "Direct Methane Conversion" which allows for the collection, filtration, and conversion of methane into graphene and hydrogen. Research was presented at the United Nations headquarters to global thought leaders, entrepreneurs, and innovators. Answered questions from a panel of judges selected from industry and academia.

Artist of the Year Finalist in Film *April 2015*

Orange County Register

Nominated by high school video teacher. Submitted "Holding Back the Tide" and became one of ten finalists. Interviewed by a panel of professional filmmakers and instructors from colleges such as University of California and Chapman.